

Let  $I = (a, f) \subseteq R[x]$  be an ideal generated by  $a, f \in R[x]$ .

Consider the diagram:

$$\begin{array}{ccccc} R[x] & \xrightarrow{\phi_a} & (R/aR)[x] & \xrightarrow{\pi_f} & (R/aR)[x]/(\overline{f}) \\ & \searrow \pi_a & \nearrow \cong & & \\ & & R[x]/aR[x] & & \end{array}$$

$$\begin{array}{ll} R[x] \rightarrow R[x]/aR[x] & R[x]/aR[x] \rightarrow (R/aR)[x] \\ p(x) \mapsto p(x) + aR[x] & p(x) + aR[x] \mapsto \overline{p(x)}, \text{ where } \overline{\sum a_n \cdot x^n} = \sum (a_n + aR) \cdot x^n \\ \text{onto Homomorphism} & \text{Isomorphism} \end{array}$$

$\Rightarrow \phi: R[x] \rightarrow (R/aR)[x]$  is onto Homomorphism, so is  $\pi_f \circ \phi$  onto Homomorphism. Moreover,

$$\begin{aligned} \ker(\pi_f \circ \phi) &= (\pi_f \circ \phi)^{-1}[\{0\}] = \phi^{-1}[\ker \pi_f] \\ &= \{p(x) \in R[x] \mid \overline{p(x)} \in \ker \pi_f\} = \{p(x) \in R[x] \mid \overline{p(x)} \in (\overline{f(x)})\} \\ (*) &= \{p(x) \in aR[x] + f(x)R[x]\} = (a) + (f) = (a, f) = I. \end{aligned}$$

Assume  $a \in R$  and  $(a) \subseteq R$  is maximal, and  $\overline{f} \in (R/aR)[x]$  is irreducible.

Then,  $R/aR$  is a field, and therefore  $(R/aR)[x]$  is p.i.d.

So,  $(\overline{f})$  is maximal, it implies that  $(R/aR)[x]/(\overline{f})$  is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x] + fR[x]) \cong (R/aR)[x]/\overline{f} \cdot (R/aR)[x]$$

Note: (\*) is obtained by: given  $p(x) \in R[x]$ ,

$$\overline{p(x)} \in (\overline{f(x)}) \Leftrightarrow \overline{p(x)} = \overline{g(x)} \cdot \overline{f(x)} \text{ for some } g(x) \in R[x]$$

$$\Leftrightarrow p(x) - g(x)f(x) \in aR[x] \text{ for some } g(x) \in R[x]$$

[Investigate  $\mathbb{Z}[x]/(6, x^2+x+1)$

1).  $(6, x^2+x+1)$

1-1).  $x^2+x+1$  is irreducible.

$x^2+x+1$  reducible  $\Rightarrow x^2+x+1 = (x+a)(x+b)$  s.t.  $ab=1$

So  $ab=1 \Rightarrow a=b=1$  or  $a=b=-1$ , but  $\pm 1$  are not roots,

1-2).  $(6, x^2+x+1) = (2, x^2+x+1) \cap (3, x^2+x+1)$

Denote  $f = x^2+x+1$ . Then,

$$(6, f) = (6) + (f) = (2)(3) + (f) = [(2) + (f)] \cdot [(3) + (f)] = (2, f) \cdot (3, f) = (2, f) \cap (3, f)$$

Thus, by CRT,

$$\mathbb{Z}[x]/(6, f) \cong \left( \mathbb{Z}[x]/(2, f) \right) \times \left( \mathbb{Z}[x]/(3, f) \right)$$

2). Consider

$$\mathbb{Z}[x] \rightarrow (\mathbb{Z}/(2))[x] \rightarrow (\mathbb{Z}/(2))[x]/(\bar{f})$$

$$\sum a_n x^n \mapsto \sum (a_n \bmod 2) x^n \mapsto \sum (a_n \bmod 2) x^n + (\bar{f})$$

which is onto Homomorphism with kernel  $(6, f)$ .

Since  $\bar{f} = \bar{x}^2 + \bar{x} + 1$  has no root in  $\mathbb{F}_2 \cong \mathbb{Z}/(2)$ , so irreducible, thus

$$\mathbb{Z}[x]/(2, f) \cong \mathbb{F}_2[x]/(\bar{x}^2 + \bar{x} + 1) \cong \mathbb{F}_{2^2}.$$

But,  $\bar{f}$  is not irreducible over  $\mathbb{Z}/(3) \cong \mathbb{F}_3$ , as  $\bar{x}^2 + \bar{x} + 1 = (\bar{x} + 2)^2$ , so

$$\mathbb{Z}[x]/(3, f) \cong \mathbb{F}_3[x]/(\bar{x}^2 + \bar{x} + 1)$$

By the division algorithm, given  $f(x) \in \mathbb{Z}[x]$ ,

$$f(x) = b(x) \cdot (x^2+x+1) + ax + b \quad \text{for some } b(x) \in \mathbb{Z}[x] \text{ and } a, b \in \mathbb{Z},$$

$$\text{So that } \mathbb{F}_3[x]/(\bar{x}^2 + \bar{x} + 1) = \{ax + b + (\bar{x}^2 + \bar{x} + 1) \mid a, b \in \mathbb{F}_3\},$$

$$(a\bar{x} + b) \cdot (c\bar{x} + d) = ac\bar{x}^2 + (ad + bc)\bar{x} + bd = (2ac + ad + bc)\bar{x} + ac + bd$$

If  $a=0$ , then  $b$  is an element of a field  $\mathbb{F}_3$ , so it is a unit unless it is zero.

$$\text{If } a=1, \text{ then } (2c + d + bc) \cdot \bar{x} + c + bd = 1$$